

Client Intelligence Platform

Operational & Technical Handover Manual

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Document Control

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Revision History

Version	Date	Author	Change Summary
1.0	June 2026	Platform Team	Initial release — full platform documentation covering all 7 pages, 6 scoring models, data requirements, roadmap, and appendices.

Distribution List

Role	Purpose
Executive Board / Commercial Director	Strategic oversight, revenue-at-risk reporting, roadmap approval
Head of Retention	Operational configuration, AM workload management, action queue oversight
Account Managers	Daily client action list, score interpretation, outcome logging
BI / Analytics Team	Data export specification, integration implementation, quality monitoring
CRM Team	CRM connector implementation, data field mapping, daily export scheduling
Product / Platform Team	Settings calibration, model validation, code deployments
Compliance / Risk	Score accountability, audit trail, data governance

SECTION 1 — EXECUTIVE SUMMARY

The OneRoyal Client Intelligence Platform is an internal retention intelligence and commercial analytics tool that scores every active client across six quantitative dimensions, automatically prioritises them into an actionable queue, and routes specific recommendations to the correct team.

Three Operational Questions Answered

Question	Score Used	Where Visible
Who is about to leave?	Retention Risk Score	Executive Dashboard · Action Center
Who is worth the most effort to save?	Commercial Value Score + Profitability Score	Action Center · Client List
What exact action should we take — and who owns it?	Priority Score → 11-Rule Decision Tree	Action Center — Recommended Action + Owner

Business Problems Solved

Problem Without Platform	Business Impact
No risk scoring across all clients	High-value clients churn undetected
No value weighting	Equal effort spent on \$500 and \$50,000 equity accounts
No book-type-aware profitability	A-Book and B-Book clients cannot be compared fairly
No systematic action routing	Retention follow-up depends on AM availability, not data

Expected Business Outcomes

Outcome	Measurable Target
Reduce undetected churn	High-risk clients identified 30–60 days before withdrawal
Increase retention call conversion	AMs call the right clients; target >40% success rate
Prioritise revenue-generating clients	Profitability-first action queue drives ROI on retention effort
Identify growth opportunity clients	Upside Potential Score surfaces clients with expansion potential
Reactivate dormant accounts	Reactivation Score identifies the best win-back candidates
Validate model predictive accuracy	Churn prediction target: >75% AUC; withdrawal: >80% AUC

Current Platform Maturity

Capability	Status
6-score client scoring engine	■ Fully implemented
Business rules engine (configurable bands)	■ Fully implemented
9-segment client matrix	■ Fully implemented
11-rule action routing	■ Fully implemented
Dark / Light theme system	■ Fully implemented
Historical snapshot database (SQLite)	■ Implemented
Outcome tracking	■ Implemented
Model validation page	■ Implemented (with simulated data)
Real CRM integration	■■ Placeholder only — requires BI team implementation
Real Holistics integration	■■ Placeholder only — requires BI team implementation
Production database	■■ SQLite (not suitable for >10k clients in production)
User authentication	■ Not implemented
Role-based access control	■ Not implemented
Automated daily refresh	■■ Manual trigger only

SECTION 2 — PLATFORM ARCHITECTURE

2.1 Application Stack

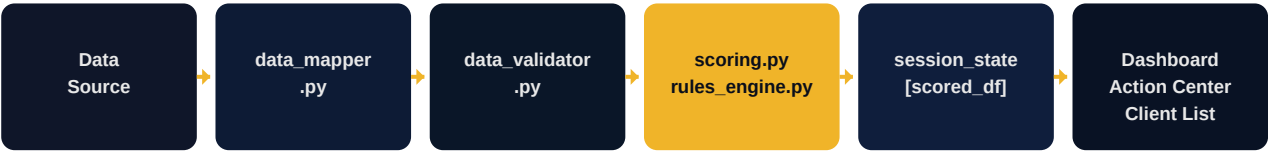
Python 3.11 + Streamlit multi-page application. Five architectural layers:

OneRoyal Client Intelligence Platform — Application Stack

PRESENTATION LAYER (pages/) 1_Executive_Dashboard · 2_Client_List · 3_Scoring_Engine
BUSINESS LOGIC LAYER (utils/) helpers.py · scoring.py · rules_engine.py
CONFIGURATION LAYER (config/) theme.py · scoring_rules.json
INTEGRATION LAYER (integrations/) data_validator.py · data_mapper.py · crm_connector.py · holistics_connector.py
DATA LAYER data/sample_data.py · data/snapshots.db (SQLite)

2.2 Data Flow

Data Flow: Source → Scoring Engine → Dashboard



Every data source (sample, file upload, CRM, Holistics) passes through the same pipeline: column renaming and date-to-days derivation in data_mapper.py, quality scoring in data_validator.py, then six-score computation by scoring.py + rules_engine.py. The resulting DataFrame is cached in st.session_state['scored_df'] for all pages.

2.3 Session State Architecture

All session variables are initialised by utils/session_init.init_session_state(), called as the first action on every page after apply_theme(). This prevents KeyError crashes when a user navigates directly to any page.

Key	Type	Default	Description
theme	str	"dark"	Current theme: "dark" or "light"
thresholds	dict	See §5	Scoring threshold values (high_risk, high_value, etc.)
risk_weights	dict	See §4.1	7 retention risk factor weights
value_weights	dict	See §4.2	7 commercial value factor weights
prof_weights	dict	See §4.3	9 profitability factor weights

Key	Type	Default	Description
react_weights	dict	See §4.4	5 reactivation factor weights
upside_weights	dict	See §4.5	5 upside potential factor weights
scoring_rules	dict	From JSON	Business rules bands (loaded from config/scoring_rules.json)
scored_df	DataFrame	Generated	Full scored client table (cached per session)
data_source	str	"sample"	Active data source key
refresh_interval	str	"manual"	Auto-refresh interval setting

2.4 Scoring Engine Architecture

The first three scores (Retention Risk, Commercial Value, Profitability) use the Business Rules Engine — configurable band scoring defined in config/scoring_rules.json. The remaining three (Reactivation, Upside Potential, Health) use direct weighted formulas in utils/scoring.py.

Score	Engine	Configurable?	Source
Retention Risk Score	Band-based (rules_engine.py)	■ Yes — via Settings	utils/rules_engine.py:score_retention_risk()
Commercial Value Score	Band-based (rules_engine.py)	■ Yes — via Settings	utils/rules_engine.py:score_commercial_value()
Profitability Score	Band-based (rules_engine.py)	■ Yes — via Settings	utils/rules_engine.py:score_profitability()
Reactivation Score	Direct weighted formula	Weights only	utils/scoring.py:score_reactivation()
Upside Potential Score	Direct weighted formula	Weights only	utils/scoring.py:compute_upside_potential()
Client Health Score	Composite formula	■ Hard-coded	utils/scoring.py:compute_health_score()
Priority Score	Composite formula	■ Hard-coded	utils/scoring.py:compute_priority_score()

SECTION 3 — DATA REQUIREMENTS

3.1 Identity Fields (Required)

Column Name	Description	Type	Re q	Example
client_id	Unique client identifier	string	■	CR10001
client_name	Full client name	string	■	Mohammed Al-Rashid
country	Country of residence	string	■	United Arab Emirates
account_manager	Assigned account manager	string	■	Sarah Johnson
ib_name	Introducing Broker name	string	Opt	Gulf Traders IB
book_type	Trading book assignment	string	■	A-Book / B-Book / M-Book
account_type	Account classification (info only)	string	Opt	Classic / Prime / Islamic
client_tenure_days	Days since account opened	integer	Opt	847

3.2 Financial Fields

Column Name	Description	Required	Example
lifetime_deposits	Total deposits ever made	■	\$45,000
withdrawals_total	Total withdrawals ever	Optional	\$12,000
net_deposits	lifetime_deposits – withdrawals_total	Derived	\$33,000
current_equity	Current account balance	■	\$28,500
equity_30d_ago	Balance 30 days ago	■ for risk	\$31,200
withdrawal_amount_last_30d	Withdrawal value in last 30 days	■ for risk	\$8,500

3.3 Revenue / Profitability Fields

These fields feed the Profitability Score and are critical for B-Book and M-Book clients. They must come from the trading platform, not the CRM.

Column Name	Description	A-Book	B-Book	M-Book
spread_commission_revenue	Monthly spread income	■	■	■
commission_revenue	Monthly commission income	■	■	■
swap_revenue	Monthly swap/overnight income	■	■	■
captured_client_losses	Net P&L; captured from B/M-Book positions	0 (zero)	■ Critical	■ Critical

Column Name	Description	A-Book	B-Book	M-Book
net_company_pnl	Monthly total profit to company (computed)	Derived	Derived	Derived

IMPORTANT — B-Book captured_client_losses: This field contains the net realised client P&L; held internally. It can be negative (when B-Book clients are profitable). The scoring engine handles negative values by assigning 0 profitability points. This field must be computed by the trading platform, not the CRM.

3.4 BI Export Specification

Recommended Lookback Periods

Metric Type	Recommended Period	Reason
Last login date	Real-time or daily snapshot	Detects sudden inactivity
Last deposit/withdrawal date	Real-time or daily snapshot	Triggers withdrawal alert rule
Trading volume	30-day rolling	Used directly in scoring
Prior period volume	Days 31–60 (previous 30-day window)	Basis for volume drop signal
Historical volume reference	90-day reference point	Used in Upside Potential Score
Current equity	Daily close balance	Core to risk and value scoring
Equity reference	30-day-ago balance	Equity trend signal in risk score
Revenue (spread, commission, swap)	Monthly aggregate	Fed into profitability score
Captured client losses (B/M)	Monthly aggregate	Profitability score, B/M book
Lifetime deposits	Cumulative from account open	Commercial value score
Complaints	30-day count	High-weight risk signal (weight ×9)

Recommended File Format

File: oneRoyal_client_intelligence_YYYYMMDD.csv | UTF-8 CSV | One row per client | Frequency: Daily at 03:00 server time | Date format: YYYY-MM-DD | Currency: USD numeric (e.g. 45000.00, not \$45,000.00) | Boolean: true/false or 1/0 | Null: empty string

Excel format: Sheet name 'Clients', Row 1 = column headers, no merged cells, no formulas, no conditional formatting. File format: .xlsx

SECTION 4 — SCORING ENGINE

All scores are on a 0–100 scale using min-max normalization: $(\text{value} - \text{min}) / (\text{max} - \text{min}) \times 100$. If all values are equal ($\text{min} = \text{max}$), the score defaults to 50. All scores are rounded to 1 decimal place.

Scoring Engine — Formula Reference

Retention Risk Score

$\Sigma \text{ weight} \times \text{band_score}(\text{signal}) \rightarrow \text{min-max } 0\text{--}100$
Signals: Withdrawal Pressure · Volume Drop · Login Inactivity · Deposit Inactivity · Complaints · Equity Erosion ·

Commercial Value Score

$\Sigma \text{ weight} \times \text{band_score}(\text{component}) \rightarrow \text{min-max } 0\text{--}100$
6 Factors: Lifetime Deposits · Net Deposits · Current Equity Volume · Redeposits · Tenure (no account type used)

Profitability Score

A-Book: spread + commission + swap
B-Book: captured_losses + commission + swap (no spread)
M-Book: $(0.6 \times \text{captured_losses}) + \text{commission} + \text{swap}$ (no spread)

Client Health Score

$(100 - \text{risk_score}) \times 0.50$
+ $\text{commercial_value_score} \times 0.30$
+ $\text{profitability_score} \times 0.20$

Priority Score

$\text{risk_score} \times 0.35$
+ $\text{commercial_value_score} \times 0.25$
+ $\text{profitability_score} \times 0.35$
+ $\text{upside_potential_score} \times 0.05$

Reactivation Score

Non-linear login window (peaks at 105 days)
+ $\text{lifetime_deposits} + \text{redeposits} + \text{vol_90d} + \text{tenure}$

4.1 Retention Risk Score

Higher score = higher risk of leaving. Source: `utils/rules_engine.py:score_retention_risk()`

Signal	Input	Band Points (Low → High)	Default Weight
Withdrawal Pressure	<code>withdrawal_30d / current_equity</code>	<10%: 10pts 10–30%: 50pts ≥30%: 90pts	12 (highest)
Volume Drop	<code>(prev_vol – curr_vol) / prev_vol</code>	≤0%: 5pts 0–30%: 30pts 30–60%: 65pts >60%: 90pts	10
Login Inactivity	<code>login_days_ago</code>	<7d: 5pts 7–30d: 25pts 30–90d: 65pts >90d: 90pts	6
Equity Trend 30d	<code>(equity_30d_ago – equity) / equity_30d_ago</code>	≤0%: 0pts 0–15%: 30pts 15–30%: 65pts >30%: 90pts	6
Deposit Inactivity	<code>last_deposit_days_ago</code>	<30d: 5pts 30–90d: 30pts 90–180d: 60pts >180d: 90pts	5
Complaints	<code>complaints_30d + open_tickets</code>	0: 0pts 1: 50pts ≥2: 90pts	5
Equity Erosion	<code>1 – (equity / net_deposits)</code>	≤0%: 0pts 0–20%: 25pts 20–50%: 60pts >50%: 90pts	5
Score Range	Label	Recommended Response	
0–29	Low Risk	Monitor — no immediate action	

Score Range	Label	Recommended Response
30–59	Medium Risk	Consider proactive contact
60–79	High Risk	Priority for retention action
80–100	Very High Risk	Immediate intervention required

4.2 Commercial Value Score

Higher score = greater commercial importance. Source: `utils/rules_engine.py:score_commercial_value()`

Component	Band Thresholds	Default Weight
Lifetime Deposits	<\$1K: 10 \$1K–\$10K: 35 \$10K–\$50K: 60 \$50K–\$200K: 85 >\$200K: 100	10 (joint highest)
Current Equity	<\$500: 5 \$500–\$5K: 30 \$5K–\$25K: 60 \$25K–\$100K: 85 >\$100K: 100	10 (joint highest)
Net Deposits	<\$0: 0 \$0–\$1K: 15 \$1K–\$10K: 40 \$10K–\$50K: 70 >\$50K: 100	8
Trading Volume (30d)	<\$1K: 5 \$1K–\$10K: 25 \$10K–\$100K: 55 \$100K–\$500K: 80 >\$500K: 100	6
Redeposit Count	0: 0 1–2: 25 3–7: 55 8–15: 80 >15: 100	6
Client Tenure	<90d: 10 90–365d: 35 365–730d: 65 >730d: 100	5

4.3 Profitability Score

Higher score = higher monthly profit to OneRoyal. Book-type-aware. Source: `utils/rules_engine.py:score_profitability()`

Book Type	Formula	Band Thresholds
A-Book	spread + commission + swap	<\$50: 10 \$50–\$200: 35 \$200–\$500: 60 \$500–\$2K: 85 >\$2K: 100
B-Book	captured_losses + commission + swap (no spread)	<\$0: 0 \$0–\$100: 20 \$100–\$500: 45 \$500–\$2K: 70 \$2K–\$5K: 90 >\$5K: 100
M-Book	$(0.6 \times \text{captured_losses}) + \text{commission} + \text{swap}$ (no spread)	<\$0: 0 \$0–\$75: 20 \$75–\$300: 45 \$300–\$1K: 70 >\$1K: 100

M-Book `internal_ratio` (default 0.6) is configurable in Settings → Business Rules Engine → Profitability Bands.

4.4 Client Health Score

Composite positive metric. Hard-coded in `utils/scoring.py:compute_health_score()`. NOT configurable via Settings.

```
health_score = (100 - risk_score) × 0.50 + commercial_value_score × 0.30 + profitability_score × 0.20
```

Range	Label	Color
80–100	Excellent	#10B981 (Green)
60–79	Healthy	#22C55E (Light Green)
40–59	Watchlist	#F59E0B (Amber)

Range	Label	Color
20–39	At Risk	#F97316 (Orange)
0–19	Critical	#EF4444 (Red)

4.5 Priority Score

Used to rank the Action Center queue. Hard-coded in `utils/scoring.py:compute_priority_score()`.

```
priority_score = risk_score × 0.35 + value_score × 0.25 + profitability_score × 0.35 +  
upside_potential_score × 0.05
```

Design rationale: Risk and profitability are equally weighted at 35% each — a client that is both at risk AND profitable generates the highest urgency. Value contributes 25%, upside potential 5%.

4.6 Segmentation Matrix & Action Routing

9-Segment Client Matrix

Source: `utils/scoring.py:assign_segments()`. Clients are tiered into Low/Medium/High for both Risk and Value using the threshold values from Settings. The 3×3 matrix produces nine named segments.

9-Segment Client Matrix: Risk × Value

	High Value	Medium Value	Low Value
High Risk	Save Immediately	Senior Retention Review	Automated Retention
Medium Risk	Proactive Nurture	Standard Nurture	Light Touch
Low Risk	High Value Growth	Growth Program	Monitor

9-Rule Action Decision Tree

Source: `utils/scoring.py:assign_recommended_action()`. First-match wins. All threshold values (`high_risk`, `high_value`, `high_profitability`, `large_withdrawal_pct`) are configurable in Settings.

Action Routing — 9-Rule Decision Tree (First Match Wins)

#	Condition	Assigned Action	Owner
1	High Risk + High Value + High Profitability	Immediate Retention Call	Management Review
2	≥2 Complaints + High Risk	Resolve Complaints + Retention Review	Retention Team
3	Large Withdrawal + High Risk	Retention Call: Withdrawal Alert	Account Manager
4	High Risk (any)	Retention Follow-Up	Retention Team
5	Upside Potential ≥65 + Value ≥70%HV	Growth Opportunity: Upgrade Offer	Sales Team
6	Reactivation ≥65 + Dormant	Reactivation Campaign	Sales Team
7	Volume Drop >50% + Risk ≥35	Re-engagement: Trading Incentive	Account Manager
8	≥1 Complaint	Complaint Resolution	Account Manager
9	Default	Monitor Only	Account Manager

Recommended Owner Routing

Condition	Assigned Owner Team
Immediate Retention Call (critical priority)	Management Review
Complaint Resolution + Retention Review / Follow-Up	Retention Team
Growth Opportunity / Reactivation / Re-engagement	Sales Team
All other actions	Account Manager

SECTION 5 — SETTINGS PAGE REFERENCE

Page: pages/5_Settings.py

5.1 Scoring Thresholds

These thresholds control when clients are labelled and which action rules trigger. They affect the segmentation matrix, action routing, and all dashboard KPIs.

Setting	Default	Min	Max	Impact of Increase	Impact of Decrease
high_risk	60	10	95	Fewer high-risk clients; team focuses on severe cases	More flagged; higher workload, more false positives
high_value	60	10	95	Fewer in 'high value' bucket; higher bar for premium treatment	More clients receive premium attention
high_profitability	60	10	95	Fewer profitability-dependent action rules trigger	More clients receive profitability-linked actions
critical_priority	65	10	95	Fewer 'Critical' labels in Action Center	More Critical labels; creates urgency inflation

5.2 Activity Thresholds

Setting	Default	Description
login_inactivity_days	30 days	Days without login before a client is flagged as inactive in action rules
large_withdrawal_pct	30% of equity	Withdrawal exceeding this % of equity triggers 'Retention Call: Withdrawal Alert'
dormant_days	30 days	Days without login before account_status changes to 'Dormant'

5.3 Business Rules Engine

Located in the lower section of Settings. Three tabs: Retention Risk Factors, Commercial Value Factors, Profitability Bands. For each factor the user can edit the Points (0–100) for each tier band. Higher points = stronger signal for that condition. After editing, clicking 'Save Scoring Rules & Rescore All Clients' writes changes to config/scoring_rules.json and immediately rescores all clients.

CAUTION: Changing band points is a calibration decision that affects every client score. Changes should be made based on validated outcome data (see Section 9), not intuition. Keep a backup of config/scoring_rules.json before making changes.

SECTION 6 — EXECUTIVE DASHBOARD

Page: pages/1_Executive_Dashboard.py

6.1 KPI Cards

KPI	Formula	Business Interpretation
Total Clients	<code>len(df)</code>	Portfolio size
Clients At Risk	<code>count(risk_score ≥ high_risk)</code>	Immediate retention workload
High-Value At Risk	<code>count(risk ≥ high_risk AND value ≥ high_value)</code>	Revenue-priority intervention candidates
Total Equity At Risk	<code>sum(current_equity where risk ≥ high_risk)</code>	Dollar value in jeopardy
Annual Revenue At Risk	<code>sum(spread + commission + swap where risk ≥ high_risk) × 12</code>	Forward revenue exposure
Annual Profitability At Risk	<code>sum(net_company_pnl where risk ≥ high_risk) × 12</code>	Forward profitability exposure

6.2 Charts

Chart	Type	Data
Retention Risk Distribution	Histogram (25 bins)	<code>retention_risk_score</code> for all clients
Client Health Distribution	Bar chart	Count per health label (5 labels)
Profitability Score Distribution	Histogram (25 bins)	<code>profitability_score</code> for all clients
Avg Profitability by Book Type	Grouped bar	Average profitability score per book type
Annual Revenue At Risk by Country	Horizontal bar (top 10)	$(\text{spread} + \text{commission} + \text{swap}) \times 12$ for at-risk clients
Annual Revenue At Risk by AM	Horizontal bar	Revenue at risk per account manager
Segmentation Matrix	3×3 Heatmap	Client count per Risk × Value cell
Trend Analytics	Multi-line (4 periods)	<code>avg_risk</code> , <code>avg_health</code> , <code>total_equity</code> , <code>annual_revenue</code>

SECTION 7 — CLIENT LIST

Page: pages/2_Client_List.py

7.1 Filter Controls

Filter	Type	Column
Search name or client ID	Text input	client_name OR client_id (case-insensitive)
Country	Dropdown	country
Account Manager	Dropdown	account_manager
IB Name	Dropdown	ib_name
Book Type	Dropdown	book_type (A-Book / B-Book / M-Book)
Risk Level	Dropdown	risk_level (Low / Medium / High)
Health	Dropdown	health_label

All filters are cumulative (AND logic). Sort column supports all 6 scores plus current_equity and lifetime_deposits.

7.2 Client Detail Panel

Below the table, a client selector shows: 6 score tiles (Risk, Value, Profitability, Reactivation, Upside Potential, Health) · Segment / Action / Owner / Reason · Financial summary (equity, deposits, PnL, account status, tenure) · Activity summary (login, deposit, withdrawal timing, volume, complaints) · Score Contribution Analysis — two horizontal bar charts showing which factors drove the Risk score and Value score.

SECTION 8 — ACTION CENTER

Page: pages/4_Action_Center.py

8.1 Prioritisation Methodology

The Action Center displays all clients except those assigned 'Monitor Only'. Sort order: Profitability Score DESC → Commercial Value Score DESC → Retention Risk Score DESC. Design rationale: protecting profitable clients is the primary commercial objective.

8.2 KPI Summary

KPI	Calculation
Clients Needing Action	Count of clients with action ≠ 'Monitor Only'
Avg Priority Score	Mean priority_score for all action clients
Annual Revenue at Stake	sum(spread + commission + swap) × 12 for action clients
Annual Profitability at Stake	sum(net_company_pnl) × 12 for action clients

8.3 Recommended AM Daily Workflow

1. Open Action Center each morning.
2. Filter by own name under 'Filter by owner → Account Manager'.
3. Work down the list in displayed order (profitability-first).
4. For 'Immediate Retention Call' or 'Retention Follow-Up': review Client Detail page for score reasons, note the action reason text for talking points, log outcome in Model Validation.
5. Flag 'Immediate Retention Call' clients for same-day escalation to management.

Escalation path: Management Review → Head of Retention + Commercial Director. Retention Team → Retention specialists, not AMs. Sales Team → Growth and reactivation campaigns.

SECTION 9 — MODEL VALIDATION

Page: pages/7_Model_Validation.py

Tab	Content
Tab 1 — Prediction Accuracy	AUC-style accuracy metrics for churn, withdrawal, reactivation predictions. Confusion matrix. Precision/recall/lift vs. random baseline.
Tab 2 — Score Factor Breakdown	Factor-by-factor contribution analysis for Risk and Value scores across the portfolio. Box plots of score distributions.
Tab 3 — Retention Effectiveness	Outcome logging form. Success rate tracking by action type and AM. Conversion funnel.
Tab 4 — Score Trend Analysis	Historical score drift charts using snapshot_db.py data. Alerts for significant score movements.
Tab 5 — Data Snapshots	Snapshot database explorer. View stored score snapshots and outcome records. Export capability.

9.1 Prediction Accuracy Methodology

When historical snapshot data and outcome data exist in snapshots.db, the platform computes an AUC-style metric by: (1) Loading score_snapshots (daily risk scores for all clients), (2) Joining with client_outcomes on client_id + outcome_type, (3) Sorting clients by risk score DESC and computing what fraction of actual churners appear in the top X% of the ranked list, (4) Computing the area under that detection curve normalised to 0–100%.

9.2 Target Accuracy Levels

Metric	Target	Current Status
Churn Prediction AUC	> 75%	Simulated — requires real outcome data to validate
Withdrawal Prediction AUC	> 80%	Simulated — requires real outcome data to validate
Reactivation Prediction AUC	> 65%	Simulated — requires real outcome data to validate

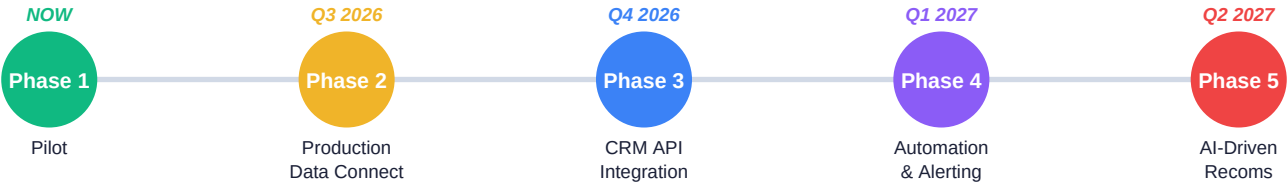
9.3 Quarterly Model Review Framework

Review Item	Responsible	Frequency
Churn AUC vs. target	Analytics Team	Quarterly
Factor correlation check (any factor decorrelated from outcomes?)	Analytics Team	Quarterly
Threshold calibration review (high_risk, high_value)	Head of Retention	Quarterly
Band point audit (scoring_rules.json vs. actual signal quality)	Product Team	Bi-annually
Retention action success rate (log in Model Validation)	Head of Retention	Monthly

Review Item	Responsible	Frequency
AM outcome logging compliance check	Head of Retention	Monthly

SECTION 10 — IMPLEMENTATION ROADMAP

Implementation Roadmap — 5 Phases



Phase	Title	Timeline	Key Deliverables	Status
1	Pilot	NOW	Synthetic data, all 7 pages, scoring engine, settings	■ Complete
2	Production Data Connect	Q3 2026	BI team exports real CSV daily, data_mapper.py updated for real column names, data_validator.py quality gates tuned, first real client scores	■■ Pending BI team
3	CRM API Integration	Q4 2026	crm_connector.py implemented (CRM API endpoint, auth), holistics_connector.py implemented, manual upload removed from production	■■ Pending CRM team
4	Automation & Alerting	Q1 2027	Daily auto-refresh scheduler, email/Slack alerts for critical risk events, PostgreSQL migration for production scale	■ Not started
5	AI-Driven Recommendations	Q2 2027	ML churn model replacing rules-based risk score, NLP action recommendation using client communication history	■ Not started

SECTION 11 — GOVERNANCE

11.1 Role Definitions

Role	Responsibilities in Platform
Head of Retention	Owns threshold and weight calibration; reviews model accuracy quarterly; approves changes to action routing rules
Account Managers	Daily action queue execution; outcome logging in Model Validation; flag data quality issues
BI / Analytics Team	Daily CSV export to correct schema; maintains data pipeline; monitors data quality score; implements Holistics connector
CRM Team	Implements <code>crm_connector.py</code> against actual CRM API; ensures field mapping; manages authentication
Platform / Product Team	Code deployments; Settings calibration support; <code>rules_engine</code> tuning; model validation interpretation
Commercial Director	Executive Dashboard consumer; approves roadmap phases; reviews revenue-at-risk reports
Compliance / Risk	Score accountability; audit trail review; GDPR compliance for client data in <code>snapshots.db</code>

11.2 Data Quality Controls

Control	Threshold	Action if Breached
Missing required fields (<code>client_id</code> , <code>book_type</code> , <code>current_equity</code>)	Any missing = error	Block upload; reject data source
Data quality score (computed by <code>data_validator.py</code>)	< 70 = warning, < 50 = fail	Display warning; refuse to rescore if fail
Duplicate client IDs	Any duplicate = error	Flag and deduplicate on most recent record
<code>current_equity</code> negative values	Any negative	Set to 0; flag in quality report
Stale data (<code>last_login_date</code> > 365 days in future)	Any future date	Flag as data error; use None

SECTION 12 — OPERATING MANUAL

12.1 First-Time Setup

1. Clone repository: `git clone [repo_url]`
2. Install dependencies: `pip install -r requirements.txt`
3. Run application: `streamlit run app.py`
4. Navigate to Settings (page 5) and verify default thresholds are appropriate for your portfolio size.
5. Navigate to Scoring Engine (page 3) to review default weights.
6. Review sample data in Client List (page 2) to understand the scoring output format.

12.2 Uploading Real Client Data

1. Prepare your data file matching the BI Export Specification in Section 3.4.
2. Navigate to Data Management (page 6).
3. Select 'Upload File' as the data source.
4. Upload the CSV or Excel file.
5. Review the Data Quality Report — ensure quality score > 70.
6. Click 'Rescore All Clients'.
7. Verify the Executive Dashboard shows the expected client count and score distributions.

12.3 Daily Analysis Workflow

Step	Who	Page	Action
1	AM / Retention Lead	Action Center (4)	Review action queue sorted by profitability. Filter by own name.
2	AM	Client List (2)	Click through any 'Immediate Retention Call' clients for score detail.
3	AM	Model Validation (7)	Log outcome of any retention actions taken yesterday.
4	Head of Retention	Executive Dashboard (1)	Check KPI cards and 'High-Value At Risk' count.
5	BI Team	Data Management (6)	Upload fresh daily export. Verify quality score > 70.
6	Analytics (weekly)	Model Validation (7)	Review prediction accuracy trends and score drift.

12.4 Interpreting Scores

Score	What It Means	Action Trigger
Retention Risk $\geq$ 60	Client shows 2+ warning signals	Call within 24 hours
Commercial Value $\geq$ 60	Client is in top third for revenue and relationship value	Prioritise over lower-value clients

Score	What It Means	Action Trigger
Profitability $\geq$ 60	Client is a significant monthly profit contributor	Assign to senior AM or retention specialist
Health = Critical (0–19)	High risk, low value, low profitability	Automated retention or deprioritise
Priority $\geq$ 65	Critical priority label — top action urgency	Escalate to team lead
Reactivation $\geq$ 65	Dormant client with high win-back probability	Launch reactivation campaign
Upside Potential $\geq$ 65	Client with strong growth indicators (equity, volume, loyalty)	Initiate upgrade offer conversation

12.5 Adjusting Settings

Threshold changes affect every labelled metric in the platform instantly. Before changing thresholds: (1) note current client counts in the Impact Preview section of Settings, (2) make the change, (3) click 'Save Settings & Rescore All Clients', (4) review the Impact Preview to verify the new client counts are operationally manageable.

Typical calibration: if the Action Center shows >30% of clients needing action, raise `high_risk` by 5–10 points. If fewer than 5% are flagged, lower it by 5 points.

SECTION 13 — APPENDICES

Appendix A — Complete Data Dictionary

Column	Data Type	Source	Required	Scoring Impact
client_id	string	CRM	■	Join key — no scoring impact
client_name	string	CRM	■	Display only
country	string	CRM	■	Display / filter only
account_manager	string	CRM	■	Action routing / workload
book_type	string	CRM/Platform	■	Profitability formula selector
account_type	string	CRM	Optional	Informational only — not used in scoring
client_tenure_days	integer	Derived	Optional	Value score (weight 5) + upside potential score
lifetime_deposits	float	CRM	■	Value score component (weight 8)
withdrawals_total	float	CRM	Optional	net_deposits derivation
net_deposits	float	Derived	Derived	Value score + equity erosion risk signal
current_equity	float	Platform	■	Risk score, value score (weight 8), withdrawal alert
equity_30d_ago	float	Platform	■ for risk	Equity trend risk signal (weight 6)
login_days_ago	integer	CRM (derived)	■	Risk score (weight 6) + reactivation score
last_deposit_days_ago	integer	CRM (derived)	■	Risk score (weight 5)
withdrawal_amount_last_30d	float	Platform	■	Risk score withdrawal pressure (weight 8)
trading_volume_last_30d	float	Platform	■	Risk score volume drop + value score
trading_volume_previous_30d	float	Platform	■ for risk	Volume drop risk signal (weight 7)
volume_90d_ago	float	Platform	Optional	Upside potential volume trend + reactivation score
spread_commission_revenue	float	Platform	■	A-Book profitability only (fee-based)
commission_revenue	float	Platform	Optional	All three profitability book formulas
swap_revenue	float	Platform	Optional	All three profitability book formulas
captured_client_losses	float	Platform	B/M-Book only	B-Book and M-Book profitability formulas
net_company_pnl	float	Derived	Derived	Priority score, Action Center ranking
number_of_redeposits	integer	CRM/Platform	Optional	Value score (weight 6) + reactivation + upside potential

Column	Data Type	Source	Required	Scoring Impact
complaints_last_30d	integer	CRM	■	Risk score (weight 9) — highest single weight
open_tickets	integer	CRM	Optional	Added to complaints for risk signal

Appendix B — Formula Dictionary

Metric	Formula	Notes
Withdrawal Pressure	$\text{withdrawal_30d} / \text{current_equity}$	Clamped 0–1
Volume Drop	$(\text{prev_vol} - \text{curr_vol}) / \text{prev_vol}$	Negative = volume growing (treated as 0 risk)
Equity Erosion	$1 - (\text{current_equity} / \text{net_deposits})$	Negative net_deposits → set to 0
Equity Trend 30d	$(\text{equity_30d_ago} - \text{equity}) / \text{equity_30d_ago}$	Negative = equity growing (treated as 0 risk)
Net Deposits	$\text{lifetime_deposits} - \text{withdrawals_total}$	Can be negative
A-Book Profitability	$\text{spread} + \text{commission} + \text{swap}$	Pure fee-based
B-Book Profitability	$\text{captured_losses} + \text{commission} + \text{swap}$ (no spread)	Negative when client profitable
M-Book Profitability	$(0.6 \times \text{captured_losses}) + \text{commission} + \text{swap}$ (no spread)	Ratio configurable in Settings
Health Score	$(100 - \text{risk}) \times 0.50 + \text{value} \times 0.30 + \text{profit} \times 0.20$	Hard-coded weights
Priority Score	$\text{risk} \times 0.35 + \text{value} \times 0.25 + \text{profit} \times 0.35 + \text{upside} \times 0.05$	Hard-coded weights
Annual Revenue (at risk)	$\text{sum}(\text{spread} + \text{commission} + \text{swap where at risk}) \times 12$	Approximation based on monthly data
Min-Max Normalisation	$(x - \text{min}) / (\text{max} - \text{min}) \times 100$	Defaults to 50 if min = max
Reactivation Login Curve	Peaks at 105 days; non-linear curve defined in scoring.py	Only meaningful for dormant clients

Appendix C — BI Export Column Mapping

This table maps the recommended BI export column names to the application's internal schema. The data_mapper.py file performs this translation on upload.

BI Export Column	Internal Column	Transformation Required
client_id	client_id	None
client_name	client_name	None
country	country	None
account_manager	account_manager	None
book_type	book_type	None
account_type	account_type	Informational — not scored

BI Export Column	Internal Column	Transformation Required
last_login_date	login_days_ago	DATE → (today – date).days
last_deposit_date	last_deposit_days_ago	DATE → (today – date).days
last_withdrawal_date	last_withdrawal_days_ago	DATE → (today – date).days
withdrawals_30d	withdrawal_amount_last_30d	Rename only
trading_volume_30d	trading_volume_last_30d	Rename only
trading_volume_previous_30d	trading_volume_previous_30d	None
volume_90d_ago	volume_90d_ago	None
spread_revenue	spread_commission_revenue	Rename only
commission_revenue	commission_revenue	None
swap_revenue	swap_revenue	None
company_pnl	net_company_pnl	Rename only
captured_client_losses	captured_client_losses	None
number_of_redeposits	number_of_redeposits	None
complaints	complaints_last_30d	Rename only
open_tickets	open_tickets	None
client_tenure_months	client_tenure_days	months × 30.44

Appendix D — Recommended CRM Integration Design

The CRM connector (integrations/crm_connector.py) currently contains placeholder methods with NotImplementedError. The BI team must implement the following:

Method	Expected Return	Implementation Notes
connect()	bool	Authenticate against CRM API. Store session token.
fetch_clients()	pd.DataFrame (internal schema)	HTTP GET against CRM clients endpoint. Map columns using data_mapper.py.
test_connection()	dict with 'success', 'client_count'	Lightweight ping to verify connectivity.
get_client(client_id)	dict (single client record)	GET single client for real-time refresh.

Minimum viable CRM API requirements: REST or GraphQL endpoint returning all active client accounts with the fields listed in Appendix C. Authentication: API key or OAuth2. Response format: JSON array or CSV download. Rate limit: must support full portfolio refresh within 2 minutes. Recommended: daily batch export to file rather than real-time API pull for large portfolios (>1,000 clients).

Appendix E — Future Enhancement Roadmap

Enhancement	Phase	Expected Benefit
PostgreSQL migration	Phase 4	Multi-user concurrency, >10k clients, proper backup
User authentication (login screen)	Phase 4	Role-based data access, audit trail by user
Email/Slack alerts on critical risk events	Phase 4	Proactive notification without manual dashboard check
ML churn model (replace rules-based risk score)	Phase 5	Higher prediction accuracy; learns from real outcomes
NLP action recommendation	Phase 5	Personalised talking points based on client communication history
Mobile-responsive UI	Phase 4	AMs can review action queue on mobile devices
Automated outcome import from CRM	Phase 3–4	Eliminate manual outcome logging; improve model feedback loop
Multi-portfolio / multi-brand support	Phase 4+	Scale to group-level analytics across multiple entities